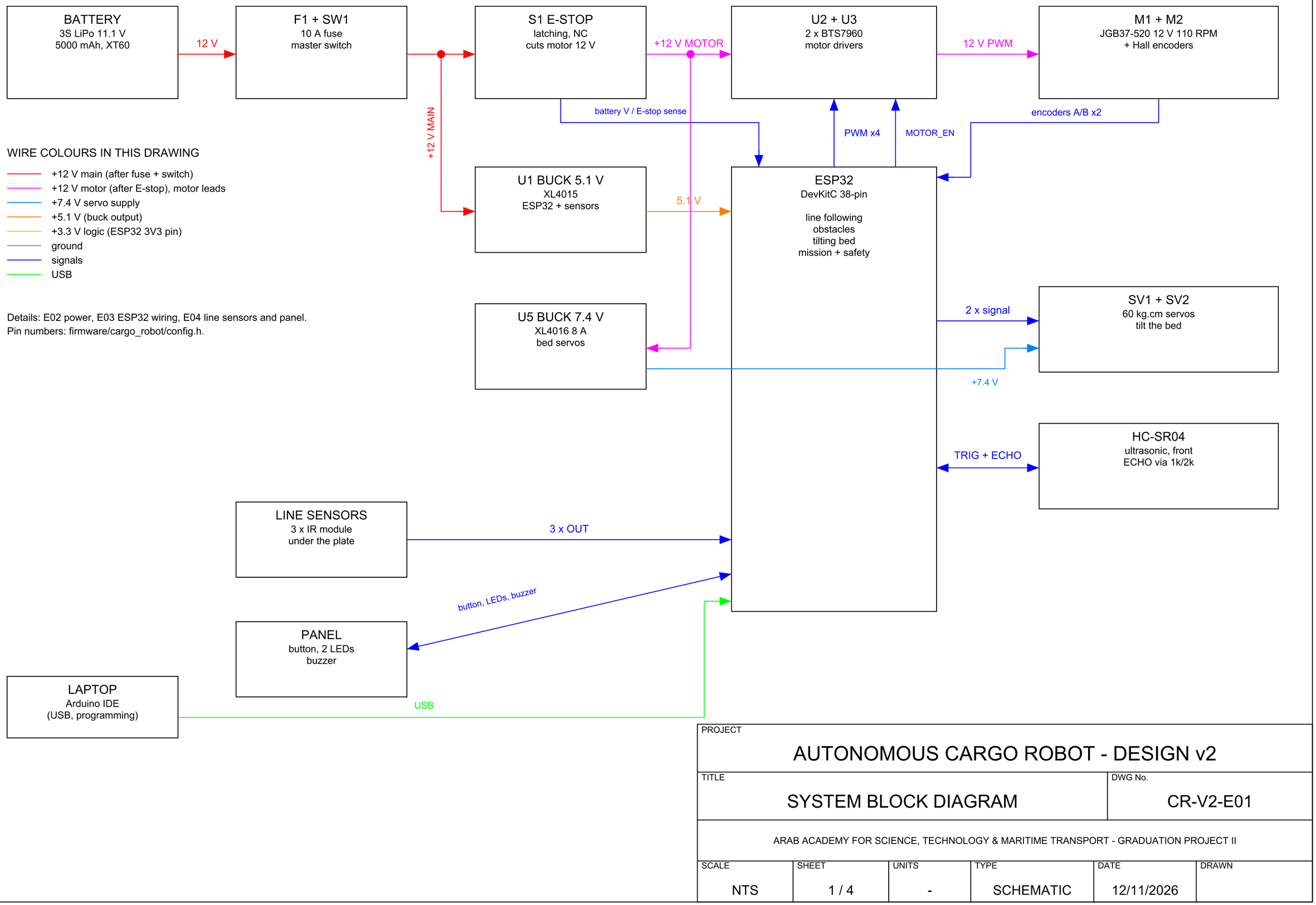
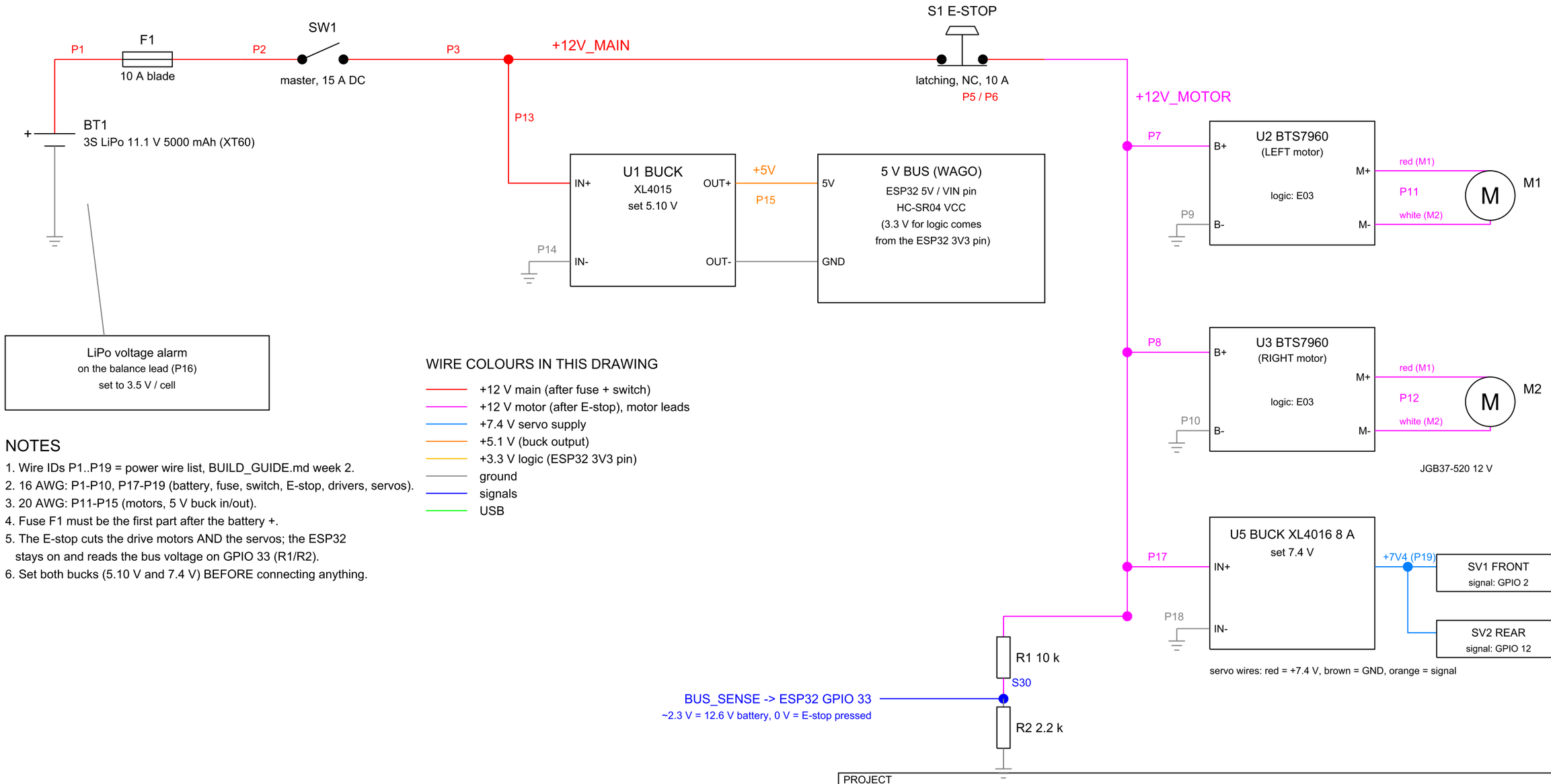


PROJECT	AUTONOMOUS CARGO ROBOT - DESIGN v2				
TITLE	SYSTEM BLOCK DIAGRAM			DWG No.	CR-V2-E01
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	TYPE	DATE	DRAWN
NTS	1 / 4	-	SCHEMATIC	12/11/2026	



WIRE COLOURS IN THIS DRAWING

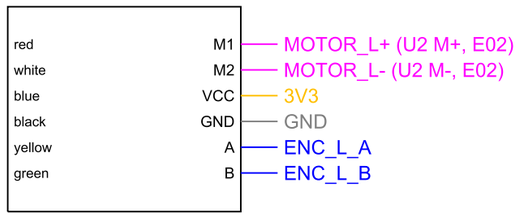
- +12 V main (after fuse + switch)
- +12 V motor (after E-stop), motor leads
- +7.4 V servo supply
- +5.1 V (buck output)
- +3.3 V logic (ESP32 3V3 pin)
- ground
- signals
- USB

NOTES

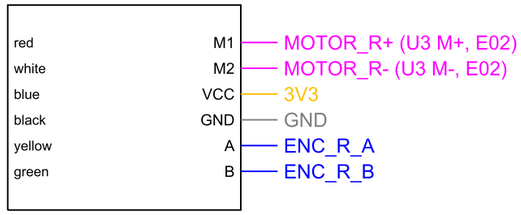
1. Wire IDs P1..P19 = power wire list, BUILD_GUIDE.md week 2.
2. 16 AWG: P1-P10, P17-P19 (battery, fuse, switch, E-stop, drivers, servos).
3. 20 AWG: P11-P15 (motors, 5 V buck in/out).
4. Fuse F1 must be the first part after the battery +.
5. The E-stop cuts the drive motors AND the servos; the ESP32 stays on and reads the bus voltage on GPIO 33 (R1/R2).
6. Set both bucks (5.10 V and 7.4 V) BEFORE connecting anything.

PROJECT	AUTONOMOUS CARGO ROBOT - DESIGN v2				
TITLE	POWER DISTRIBUTION			DWG No.	CR-V2-E02
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	TYPE	DATE	DRAWN
NTS	2 / 4	-	SCHEMATIC	12/11/2026	

M1 LEFT MOTOR CABLE



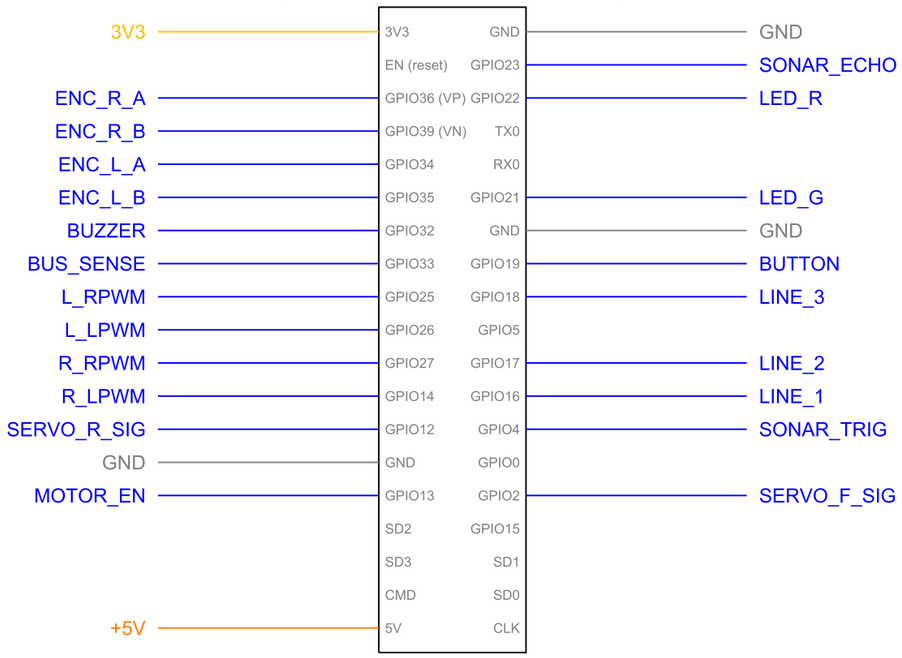
M2 RIGHT MOTOR CABLE



Encoder VCC = 3.3 V ONLY. GPIO 34/35/36/39
have no pull-ups: 10k from each ENC line to 3V3.

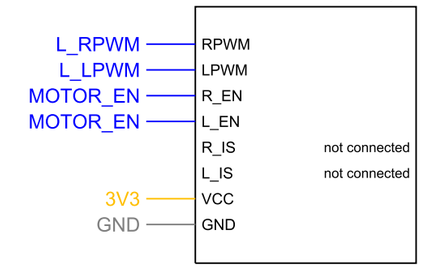
ESP32 DevKitC 38-pin

(on its screw-terminal breakout; USB end at the bottom)

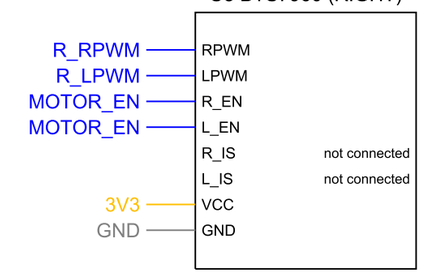


SD0-SD3, CMD, CLK: flash - never use. GPIO 0, 5, 15: free (start-up pins).
GPIO 12 must be LOW at start-up: only the rear servo signal on it.

U2 BTS7960 (LEFT)

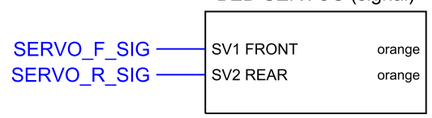


U3 BTS7960 (RIGHT)



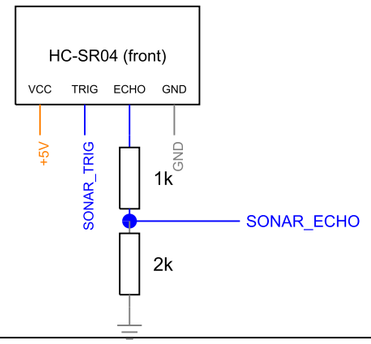
MOTOR_EN = GPIO 13, with a 10k pull-down
to GND (E04): motors off at power-up.

BED SERVOS (signal)



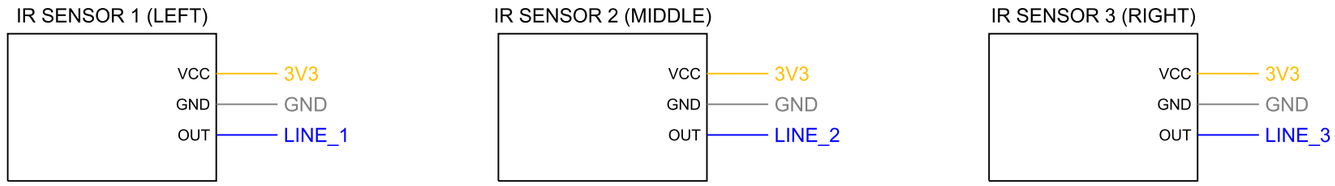
red/brown: +7V4 bus (E02)

ECHO is 5 V: the 1k / 2k divider makes it 3.3 V.



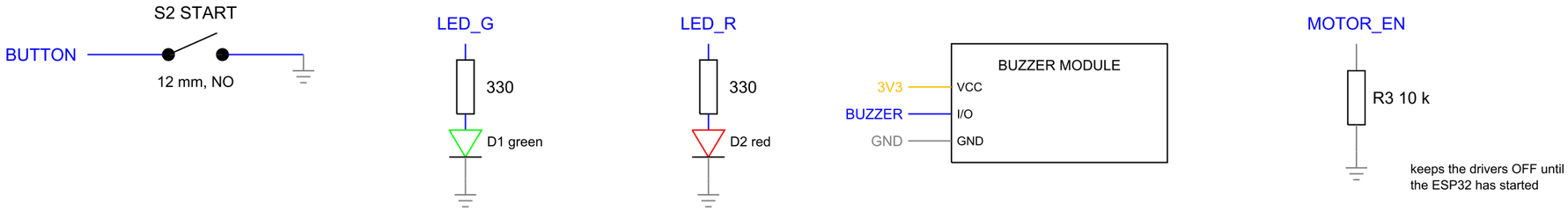
Same net name = same wire. Wire IDs S1..S28: BUILD_GUIDE.md week 3.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: ESP32 WIRING				DWG No.: CR-V2-E03	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: NTS	SHEET: 3 / 4	UNITS: -	TYPE: SCHEMATIC	DATE: 12/11/2026	DRAWN:



3-pin IR obstacle-avoidance modules (FC-51 type), on the printed bar under the bottom plate. LINE_1 = GPIO 16, LINE_2 = GPIO 17, LINE_3 = GPIO 18.
Power them from 3V3, NOT 5 V: then their OUT is safe for the ESP32. Each blue pot: set so the module's LED switches between the floor and the black tape.

PANEL (BUTTON = GPIO 19, LED_G = 21, LED_R = 22, BUZZER = 32) AND PULL-DOWN (on the I/O perfboard)



NOTES

1. All logic on this sheet runs on 3.3 V from the ESP32's 3V3 pin. No I2C and no I/O expander.
2. Line sensors: output HIGH over black tape (setting line_black). Check with the "line" command.
3. Same net name = same wire. Pin numbers: firmware/cargo_robot/config.h.

WIRE COLOURS IN THIS DRAWING

- +12 V main (after fuse + switch)
- +12 V motor (after E-stop), motor leads
- +7.4 V servo supply
- +5.1 V (buck output)
- +3.3 V logic (ESP32 3V3 pin)
- ground
- signals
- USB

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: LINE SENSORS AND PANEL				DWG No.: CR-V2-E04	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: NTS	SHEET: 4 / 4	UNITS: -	TYPE: SCHEMATIC	DATE: 12/11/2026	DRAWN: